

LEGEND

- C1

USB 3.0 Standard-B edge X-, centre 39.01, body 13.00 wide, slot -0.80 to 12.60 about the PCB top face
- C2

5 V DC barrel jack edge X-, centre 58.24, body 9.40 wide, slot -0.60 to 11.40 about the PCB top face
- C3

SMA, TX edge X+, centre 20.45, body 9.50 wide, bored 11.00 dia, axis 0.45 above the PCB top face
- C4

SMA, RX edge X+, centre 60.33, body 9.50 wide, bored 11.00 dia, axis 0.45 above the PCB top face
- C5

SMB reference clock, right angle edge Y+, centre 91.25, body 8.00 wide, bored 10.00 dia, axis 5.00 above the PCB top face
- U1

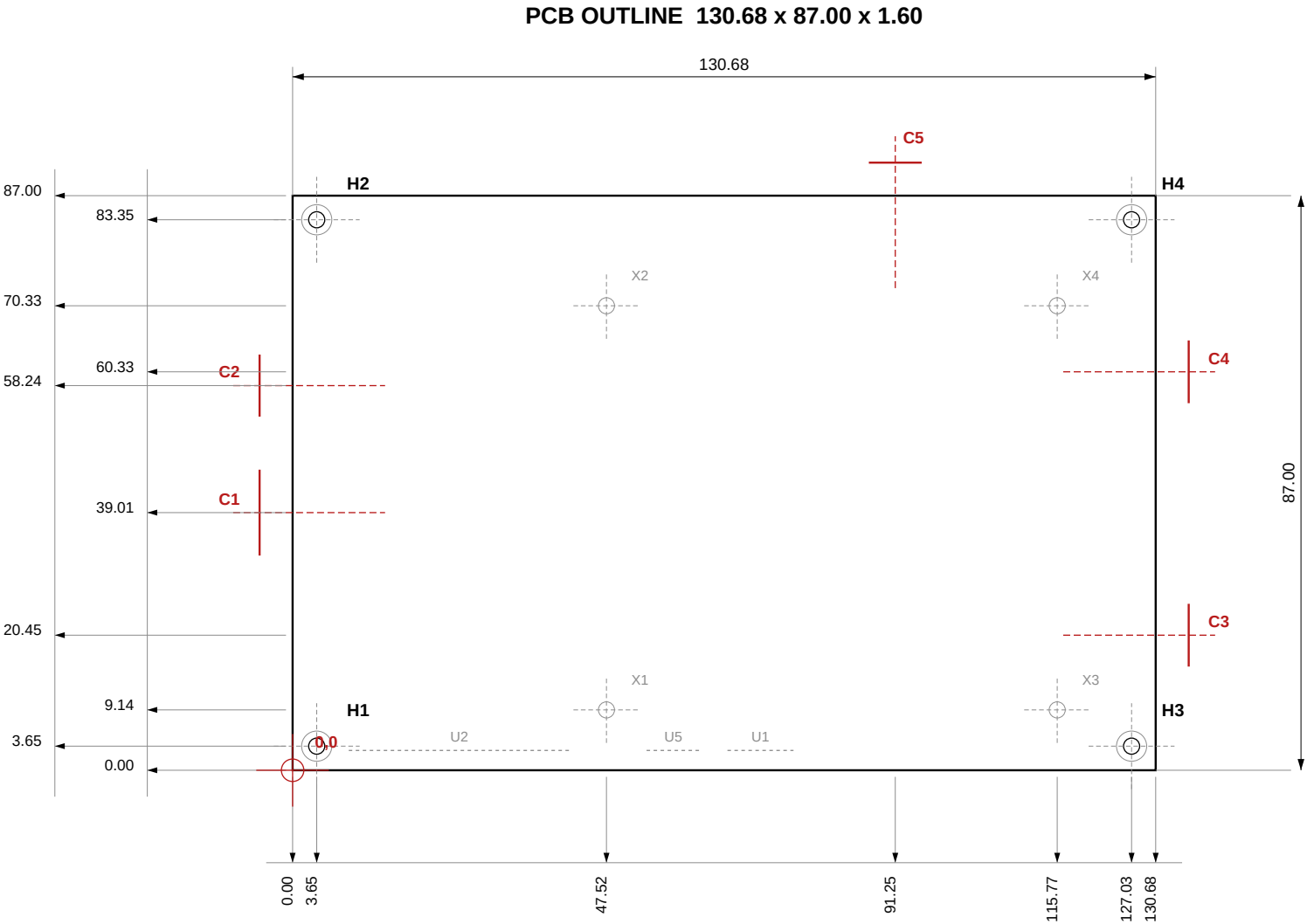
Third SMA position, unfitted edge Y-, centre 70.84
- U2

Expansion header, 2 x 20 at 1.27 mm edge Y-, centre 25.25
- U3

U.FL alternate, 2.1 mm inboard edge X+, centre 34.79
- U4

U.FL alternate, 2.1 mm inboard edge X+, centre 45.22
- U5

U.FL alternate, 1.4 mm inboard edge Y-, centre 57.66



H1 to H4 mounting holes used by this case. 2.44 drill shown solid,  
0.180 in (4.57) DXF pad shown grey. Nuand specify 2-56 screws,  
so the drill is a #2 clearance. Measure yours to confirm.

X1 to X4 further DXF holes, most likely XB-200 standoffs. Not used.

Red connector body centre and width at the board edge.

Grey features that get no wall opening.

## PCB FEATURES (datum = bottom-left PCB corner)

| Ref | Feature                             | X      | Y     | Note       |
|-----|-------------------------------------|--------|-------|------------|
| H1  | Mounting hole, case screw           | 3.65   | 3.65  | 2.44 clr   |
| H2  | Mounting hole, case screw           | 3.65   | 83.35 | 2.44 clr   |
| H3  | Mounting hole, case screw           | 127.03 | 3.65  | 2.44 clr   |
| H4  | Mounting hole, case screw           | 127.03 | 83.35 | 2.44 clr   |
| X1  | Mounting hole, unused               | 47.52  | 9.14  | XB-200     |
| X2  | Mounting hole, unused               | 47.52  | 70.33 | XB-200     |
| X3  | Mounting hole, unused               | 115.77 | 9.14  | XB-200     |
| X4  | Mounting hole, unused               | 115.77 | 70.33 | XB-200     |
| C1  | USB 3.0 Standard-B                  | 0.00   | 39.01 | body 13.00 |
| C2  | 5 V DC barrel jack                  | 0.00   | 58.24 | body 9.40  |
| C3  | SMA, TX                             | 130.68 | 20.45 | body 9.50  |
| C4  | SMA, RX                             | 130.68 | 60.33 | body 9.50  |
| C5  | SMB reference clock, right angle    | 91.25  | 87.00 | body 8.00  |
| U1  | Third SMA position, unfitted        | 70.84  | 0.00  | no opening |
| U2  | Expansion header, 2 x 20 at 1.27 mm | 25.25  | 0.00  | no opening |
| U3  | U.FL alternate, 2.1 mm inboard      | 130.68 | 34.79 | no opening |
| U4  | U.FL alternate, 2.1 mm inboard      | 130.68 | 45.22 | no opening |
| U5  | U.FL alternate, 1.4 mm inboard      | 57.66  | 0.00  | no opening |

## WALL OPENINGS (datum = outside corner of the case)

| Ref | Edge | Form  | Centre | Size       | Height range  |
|-----|------|-------|--------|------------|---------------|
| C1  | X-   | rect  | 42.81  | 14.60 wide | 7.20 to 20.60 |
| C2  | X-   | rect  | 62.04  | 11.00 wide | 7.40 to 19.40 |
| C3  | X+   | round | 24.25  | 11.00 dia  | 2.95 to 13.95 |
| C4  | X+   | round | 64.13  | 11.00 dia  | 2.95 to 13.95 |
| C5  | Y+   | round | 95.05  | 10.00 dia  | 8.00 to 18.00 |

Rectangular openings also get an outer relief pocket 2.00 wider on every side, 2.00 deep, for cable overmoulds. Bored holes get none.  
C3 and C4 are bored on the SMA block centre, 0.45 above the PCB top face: a 9.50 x 8.10 block, 4.50 up and 2.00 down off the board.

## HEIGHT STACK-UP (from outside of base floor)

|                             |       |
|-----------------------------|-------|
| Outside of floor            | 0.00  |
| Inside of floor             | 2.40  |
| Insert bore floor           | 2.40  |
| Screw tip, fully driven     | 2.80  |
| Standoff top, PCB underside | 6.40  |
| PCB top surface             | 8.00  |
| Parting plane, base to lid  | 11.00 |
| Head seat, recess floor     | 22.80 |
| Cavity ceiling              | 23.00 |
| Outside of lid              | 25.60 |

## Overall 138.28 x 94.60 x 25.60

Cavity 132.28 x 88.60, wall 3.00, corner R3.00

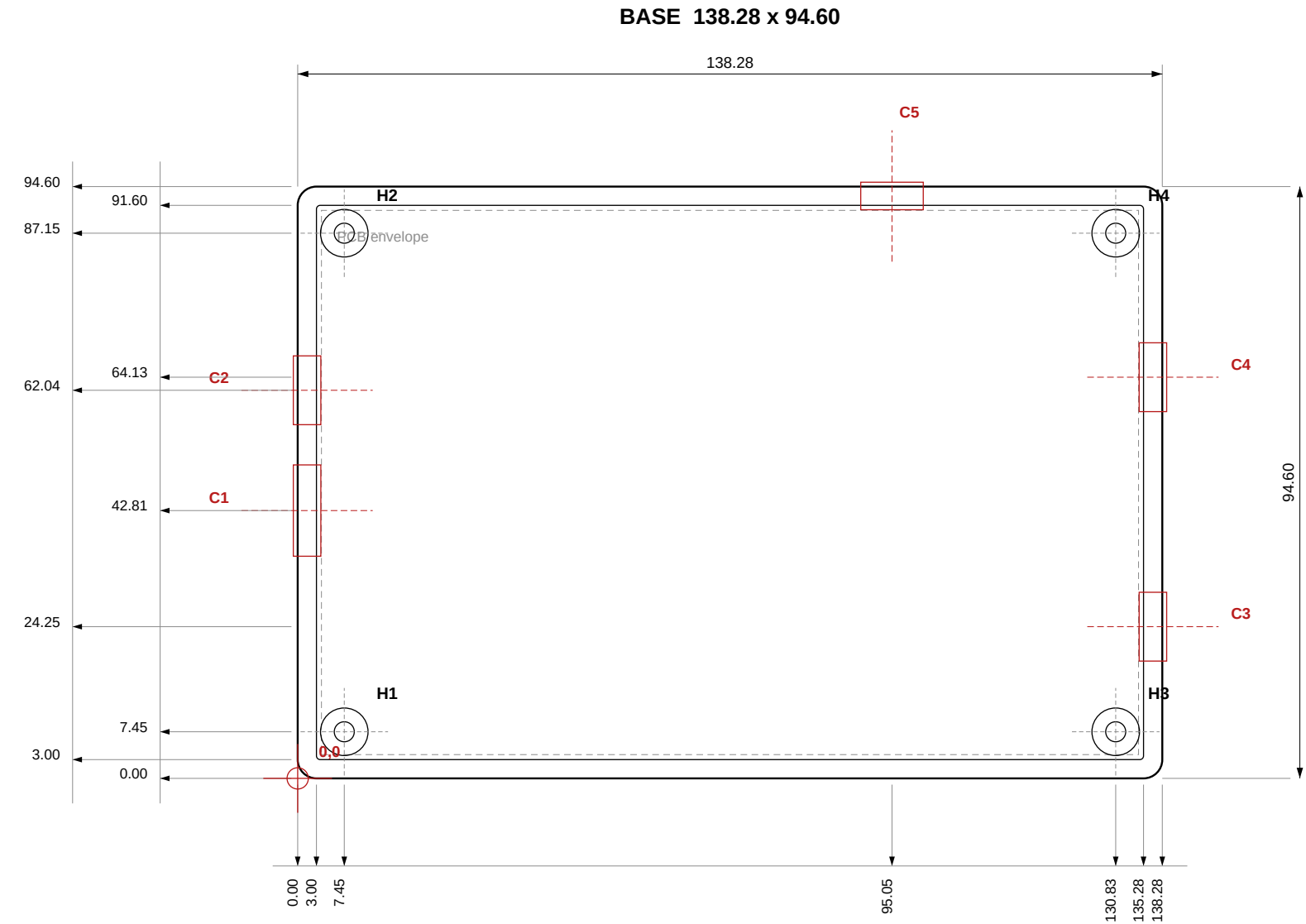
Outer edge chamfers bottom 0.80, top 1.20, both 45 deg

Fasteners 4 x M2 x 20 through lid into M2 inserts, bore 3.20 x 4.00 deep

Head recess 4.40 dia x 2.80 deep, seat z 22.80, 3.60 of thread in the insert

OPENING REFS

- C1 USB 3.0 Standard-B centre 42.81, 14.60 wide, z 7.20 to 20.60
- C2 5 V DC barrel jack centre 62.04, 11.00 wide, z 7.40 to 19.40
- C3 SMA, TX centre 24.25, 11.00 dia bore, axis z 8.45
- C4 SMA, RX centre 64.13, 11.00 dia bore, axis z 8.45
- C5 SMB reference clock, right angle centre 95.05, 10.00 dia bore, axis z 13.00

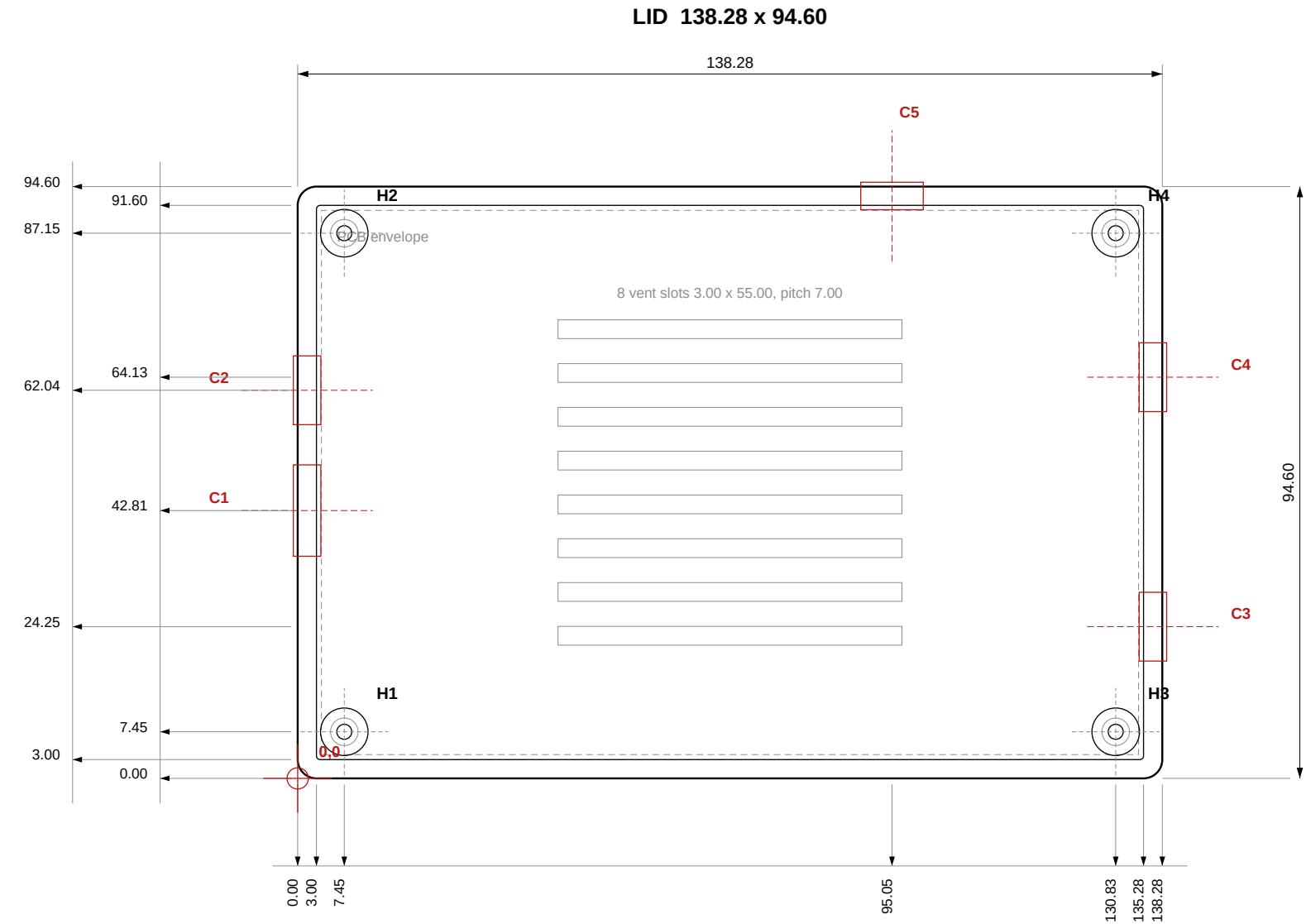


NOTES

- Standoff OD 7.60, height 4.00 above the floor.
- Insert bore 3.20 dia x 4.00 deep for M2 heat-set inserts.
- Wall reduced to 1.25 for the top 3.00 to form the lip rebate.
- Base height 11.00. Floor 2.40.
- Bottom outer edge chamfered 0.80 x 45 deg. See sheet 5.

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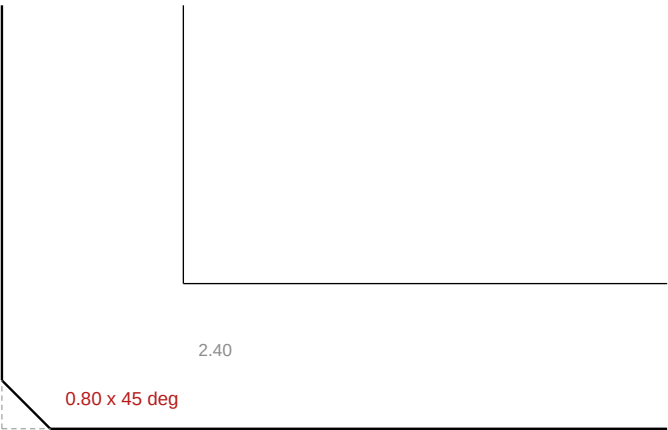


NOTES

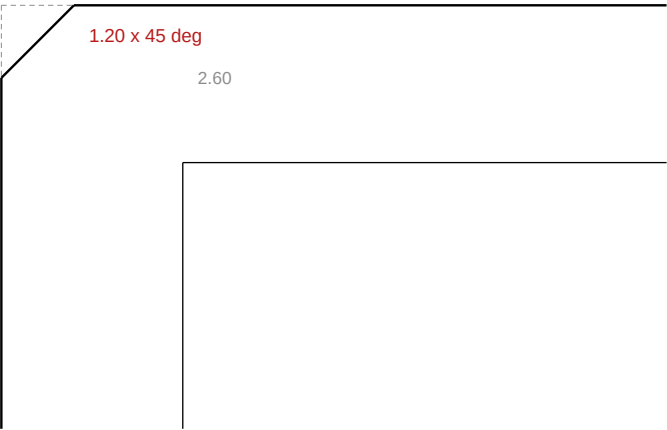
- Boss OD 7.60, from PCB top face to cavity ceiling.
- Screw 2.40 clr, head recess 4.40 dia x 2.80 deep, seat z 22.80, 45 deg cone under.
- Recess depth follows the M2 x 20 screw, not the plate thickness.
- Tip stops 0.40 above the insert bore floor, 3.60 of thread engaged.
- Lip 1.20 thick x 3.00 deep, 0.20 slip fit, 0.35 relief off the cavity wall.
- Lid height 14.60 above the parting plane. Top plate 2.60.
- Top outer edge chamfered 1.20 x 45 deg. See sheet 5.

DETAIL OUTER EDGE CHAMFERS 8:1

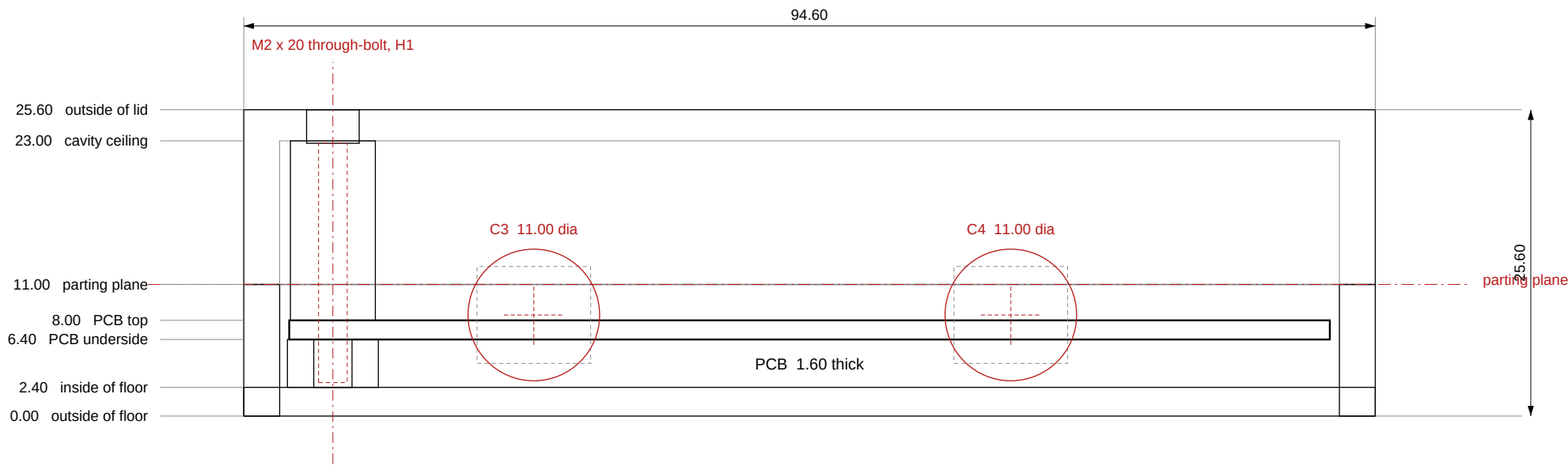
Base, bottom outer edge



Lid, top outer edge



Wall 3.00. Parting line edges left sharp so the halves mate flush.

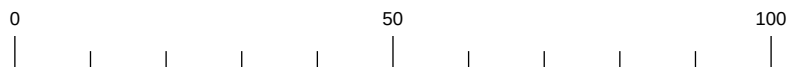
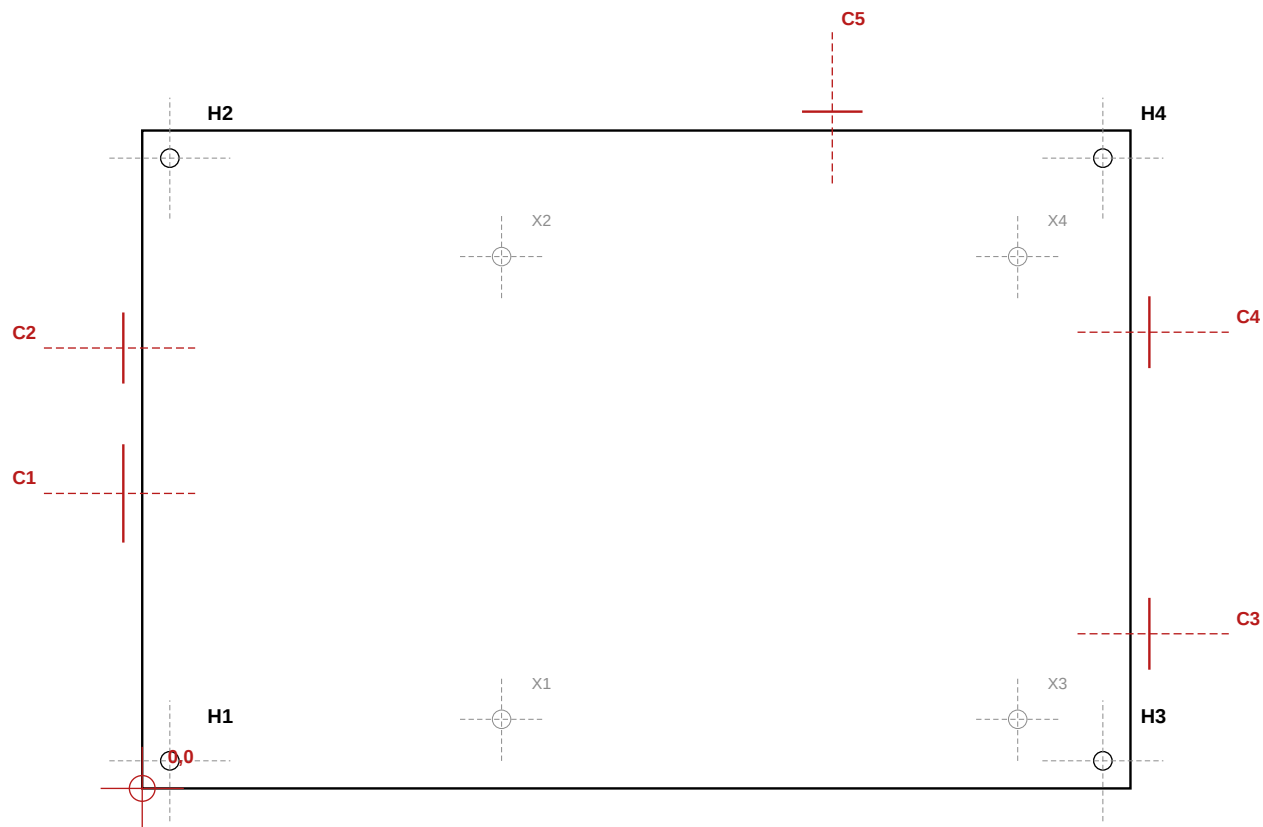


Cut plane runs parallel to the short PCB edge, 7.45 from the case datum, through openings C3 and C4 and the H1 screw column.

Drawn at 2:1. Values on the left are heights above the outside of the base floor.

M2 x 20 screw: head seat z 22.80, recess 2.80 deep, tip z 2.80, 3.60 of thread inside the 4.00 deep insert bore.

C3 and C4 bored 11.00 dia about the SMA block centre, 0.45 above the PCB top face. Grey dashed outline is the 9.50 x 8.10 block, 4.50 above and 2.00 below the board.



**SCALE CHECK. This bar must measure exactly 100.0 mm.**

If it does not, reprint with scaling set to 100 percent or actual size.

Lay the PCB on the outline. Every hole centre and connector centreline should sit under its mark.

Ticks outside the outline show connector body width; the dashed line is its centre.

**bladeRF x40 enclosure**

**PRINT AT 100 PERCENT, NO FIT TO PAGE**

All dimensions mm. Datum marked on each view.